



10300 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
PSO J318.5-22, TSO				
	1	NIRSPEC/PRISM	NIRSpec Bright Object Time Series	(1) PSOJ318.5-22
	2	MIRI/MRS	MIRI Medium Resolution Spectroscopy	(1) PSOJ318.5-22
	3	MIRI/MRS	MIRI Medium Resolution Spectroscopy	(1) PSOJ318.5-22
	4	MIRI/MRS	MIRI Medium Resolution Spectroscopy	(1) PSOJ318.5-22
	5	MIRI/MRS	MIRI Medium Resolution Spectroscopy	(1) PSOJ318.5-22
	6	MIRI/MRS	MIRI Medium Resolution Spectroscopy	(1) PSOJ318.5-22
	7	MIRI/MRS	MIRI Medium Resolution Spectroscopy	(1) PSOJ318.5-22
	8	MIRI/MRS	MIRI Medium Resolution Spectroscopy	(1) PSOJ318.5-22

ABSTRACT

For over a decade there has been debate about what exactly drives the spectral variability of brown dwarfs and imaged exoplanets. Is it clouds, chemistry, or both? JWST has allowed us to settle this debate for the brown dwarf VHS 1256 b where clouds are conclusively shown to be the driving mechanism. Now we wish to determine whether this picture also holds for fast-spinning and lower mass exoplanets at similar temperature. We propose to observe the ~ 8.5 MJup free-floating "rogue" planet PSO J318.5-22, using a combination of NIRSpec/PRISM and MIRI/MRS, to monitor its dancing spectra across 2 rotations. PSO J318.5-22 is a reliably variable object that has been spectrally monitored previously with HST and Spitzer. It is also a near-identical analog to the high-contrast exoplanets HR 8799 bcde. These observations will allow us to test if both imaged exoplanets and brown dwarfs share the same variability driving mechanism of dynamic clouds.

OBSERVING DESCRIPTION

We will observe PSO J318.5-22 across 2 of its 8.6 hour rotations using NIRSpec PRISM and MIRI MRS in order to map its spectroscopic variability.

Rotation 1: NIRSpec/PRISM. Using the BOTs template we employ SUB2048 subarray, NRS readout (for conservative data volume), 35 groups/int, and combining every 3 integrations yields a cadence of 381 seconds. This provides $S/N > 300$ from $1.2\,\mu\text{m}$ to $5.3\,\mu\text{m}$ and $S/N \sim 500$ from $2\,\mu\text{m}$ to $4\,\mu\text{m}$ for each 381 seconds of integration. We repeat this 81 times for a total of 244 integrations which covers the first rotation of PSO J318.5-22.

JWST Proposal 10300 (Created: Friday, March 13, 2026, 2:04:40PM Eastern Standard Time) - Overview

Rotation 2: MIRI/MRS. We will repetitively cycle through the three MRS gratings (SHORT, MEDIUM, LONG) 7 times across a single rotation (well below the 10 loop limited recommended by STScI). Each MIRI MRS observation will consist of 100 groups and 4 integrations and uses the FASTR1 readout. This results in resulting in a S/N >30 per wavelength bin at $8\mu\text{m}$ to $10\mu\text{m}$. One cycle takes approximately ~ 1.2 hours, allowing 7 repetitions across a single rotation of PSO J318.5-22.

All observations must happen in sequence and are non-interruptible. We do not encounter data volume warnings. Movement of the high gain antenna would likely result in a small number of unusable data but will not threaten our science goals.

Proposal 10300 - Targets - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	PSOJ318.5-22	RA: 21 14 8.0256 (318.5334400d) Dec: -22 51 35.84 (-22.85996d) Equinox: J2000	Proper Motion RA: 136.3 mas/yr Proper Motion Dec: -144.29999994263198 mas/yr Parallax: 0.0451" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Brown dwarfs, Exoplanets]					

Proposal 10300 - Observation 1 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Observation	Proposal 10300, Observation 1: NIRSPEC/PRISM										Fri Mar 13 19:04:40 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec Bright Object Time Series										
Diagnostics	(NIRSPEC/PRISM (Obs 1)) Warning (Form): Exposure Duration exceeds the limit of 10000.0 seconds. Above this limit it is possible that a High Gain Antenna move may occur during the exposure. (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	PSOJ318.5-22	RA: 21 14 8.0256 (318.5334400d) Dec: -22 51 35.84 (-22.85996d) Equinox: J2000			Proper Motion RA: 136.3 mas/yr Proper Motion Dec: -144.29999994263198 mas/yr Parallax: 0.0451" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Brown dwarfs, Exoplanets]										
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	WATA	SUB32	F140X	NRSRAPIDD6	3	1	1	0.26	272164
Template	Subarray SUB2048										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	PRISM/CLEAR	NRS	35	244	1	1	244	31037.405	272164	

Proposal 10300 - Observation 1 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Special Requirements

Time Series Observation
No Parallel Attachments

Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible

Proposal 10300 - Observation 2 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Observation	Proposal 10300, Observation 2: MIRI/MRS												Fri Mar 13 19:04:40 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	PSOJ318.5-22	RA: 21 14 8.0256 (318.5334400d) Dec: -22 51 35.84 (-22.85996d) Equinox: J2000				Proper Motion RA: 136.3 mas/yr Proper Motion Dec: -144.29999994263198 mas/yr Parallax: 0.0451" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Brown dwarfs, Exoplanets]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	272164				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		NO			FULL			Allow Auto Reorder				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	1	SHORT(A)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164

Proposal 10300 - Observation 2 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Special Requirements	Time Series Observation No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible
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Proposal 10300 - Observation 3 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Observation	Proposal 10300, Observation 3: MIRI/MRS												Fri Mar 13 19:04:40 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	PSOJ318.5-22	RA: 21 14 8.0256 (318.5334400d) Dec: -22 51 35.84 (-22.85996d) Equinox: J2000				Proper Motion RA: 136.3 mas/yr Proper Motion Dec: -144.29999994263198 mas/yr Parallax: 0.0451" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Brown dwarfs, Exoplanets]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	272164				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		NO				FULL				Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	1	SHORT(A)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164

Proposal 10300 - Observation 3 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Special Requirements	Time Series Observation No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible
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Proposal 10300 - Observation 4 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Observation	Proposal 10300, Observation 4: MIRI/MRS												Fri Mar 13 19:04:40 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	PSOJ318.5-22	RA: 21 14 8.0256 (318.5334400d) Dec: -22 51 35.84 (-22.85996d) Equinox: J2000				Proper Motion RA: 136.3 mas/yr Proper Motion Dec: -144.29999994263198 mas/yr Parallax: 0.0451" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Brown dwarfs, Exoplanets]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	272164				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		NO				FULL				Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	1	SHORT(A)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164

Proposal 10300 - Observation 4 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Special Requirements	Time Series Observation No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible
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Proposal 10300 - Observation 5 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Observation	Proposal 10300, Observation 5: MIRI/MRS												Fri Mar 13 19:04:40 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnos	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	PSOJ318.5-22	RA: 21 14 8.0256 (318.5334400d) Dec: -22 51 35.84 (-22.85996d) Equinox: J2000				Proper Motion RA: 136.3 mas/yr Proper Motion Dec: -144.29999994263198 mas/yr Parallax: 0.0451" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Brown dwarfs, Exoplanets]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	272164				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		NO			FULL			Allow Auto Reorder				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	1	SHORT(A)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164

Proposal 10300 - Observation 5 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Special Requirements	Time Series Observation No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible
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Proposal 10300 - Observation 6 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Observation	Proposal 10300, Observation 6: MIRI/MRS												Fri Mar 13 19:04:40 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	PSOJ318.5-22	RA: 21 14 8.0256 (318.5334400d) Dec: -22 51 35.84 (-22.85996d) Equinox: J2000				Proper Motion RA: 136.3 mas/yr Proper Motion Dec: -144.29999994263198 mas/yr Parallax: 0.0451" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Brown dwarfs, Exoplanets]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	272164				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		NO				FULL				Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	1	SHORT(A)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164

Proposal 10300 - Observation 6 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Special Requirements	Time Series Observation No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible
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Proposal 10300 - Observation 7 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Observation	Proposal 10300, Observation 7: MIRI/MRS												Fri Mar 13 19:04:40 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	PSOJ318.5-22	RA: 21 14 8.0256 (318.5334400d) Dec: -22 51 35.84 (-22.85996d) Equinox: J2000				Proper Motion RA: 136.3 mas/yr Proper Motion Dec: -144.29999994263198 mas/yr Parallax: 0.0451" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Brown dwarfs, Exoplanets]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	272164				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		NO			FULL			Allow Auto Reorder				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	1	SHORT(A)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164

Proposal 10300 - Observation 7 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Special Requirements	Time Series Observation No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, 6, 7, Non-interruptible
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Proposal 10300 - Observation 8 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Observation	Proposal 10300, Observation 8: MIRI/MRS												Fri Mar 13 19:04:40 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	PSOJ318.5-22	RA: 21 14 8.0256 (318.5334400d) Dec: -22 51 35.84 (-22.85996d) Equinox: J2000				Proper Motion RA: 136.3 mas/yr Proper Motion Dec: -144.29999994263198 mas/yr Parallax: 0.0451" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Brown dwarfs, Exoplanets]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	272164				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		NO				FULL				Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	1	SHORT(A)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	2	MEDIUM(B)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSLONG		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164
	3	LONG(C)	MRSSHORT		FASTR1	100	4	1	None	1	4	1118.341	272164

Proposal 10300 - Observation 8 - Dancing 1-28 micron spectra to investigate the variable nature of a rogue planet

Special Requirements	Time Series Observation No Parallel Attachments
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